

ER Diagram

Money Exchange System

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Relationships

The model contains four entities and three relationships. The relationships form a chain that follows the flow of an exchange: a *Customer* makes a *Transaction* that involves a *Currency*, and each *Currency* is quoted against another through an *ExchangeRate*.

Makes — Customer : Transaction. A *Customer* makes a **Transaction** (a currency exchange). This is a *one-to-many* association:

- one *Customer* may make *many* transactions (N);
- each *Transaction* belongs to exactly *one* customer (1).

Involves — Transaction : Currency. Each **Transaction** involves a **Currency**. A transaction always has *two* currency roles — the *base* currency the customer sells and the *quote* currency they buy — so this is modelled as two foreign keys. From each currency's point of view:

- one *Currency* may be involved in *many* transactions ($1 : N$);
- each *Transaction* involves exactly *one* base and *one* quote currency (1).

Quotes — Currency : ExchangeRate. A **Currency** is quoted against another through an **ExchangeRate**. As above, a rate has *two* currency roles (base and quote), materialised as two foreign keys:

- one *Currency* may appear in *many* exchange rates ($1 : N$);
- each *ExchangeRate* quotes exactly *one* base and *one* quote currency (1).

Design notes

- The transaction table is named `exchange_transaction` in the relational schema because `transaction` is a reserved SQL keyword.
- A transaction stores `rate_used` as a *snapshot* of the exchange rate at the moment of the trade rather than a foreign key to `exchange_rate`. Exchange rates change daily, so keeping the exact rate on the transaction preserves an accurate, historical record even if the quoted rate is later updated.
- Foreign keys (such as `customer_id` on a transaction, or the base/quote currency codes) are carried by the relationships rather than shown as attributes in the diagram.

ER Diagram

Legend: rectangles are entities, diamonds are relationships, ellipses are attributes, and underlined attributes are primary keys.

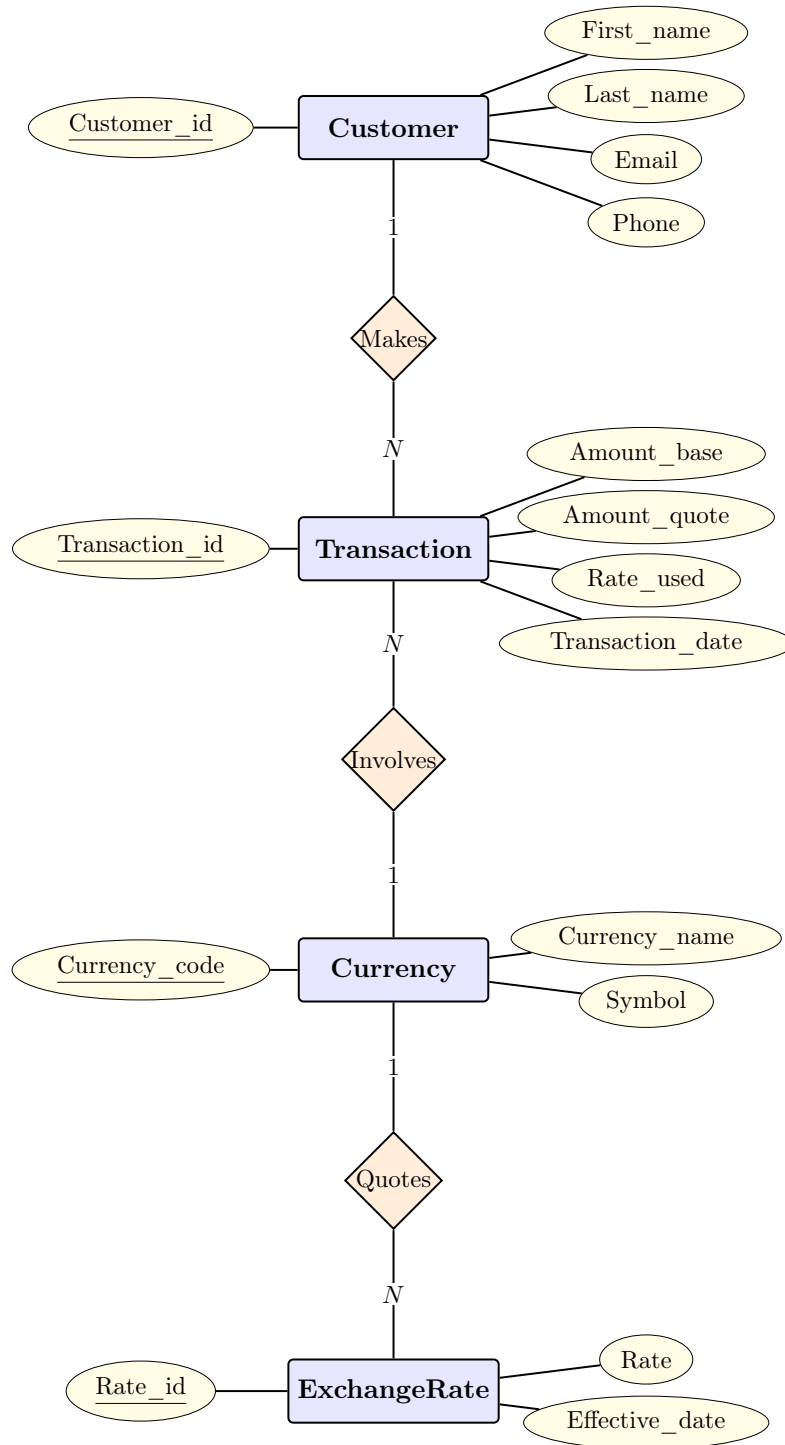


Figure 1: ER diagram in Chen notation. Primary keys are underlined. Foreign keys are carried by the relationships rather than shown as attributes.